Air Quality Analysis for Health and Environmental Protection

Project Report

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Business understanding

Air Quality Analysis for Health and Environmental Protection is a data mining project that aims to uncover the syndemics between poor air quality and respiratory virus affliction rates.

Data from the Santa Clara County Public Health Department shows that respiratory viruses like COVID-19, influenza, and RSV continuously cause serious illness, hospitalizations, and deaths in Santa Clara County [6]. California itself experienced an unusually severe flu season last winter when hospitalizations rose and led to concerns that outbreaks may last for weeks [7]. Meanwhile, investigational studies have shown that various types of air pollution can adversely affect proper immune functioning and contribute to the breakout of infectious diseases [8].

This research identifies patterns in air quality and respiratory virus datasets so it can facilitate the forecasting of pollution related health impacts. These findings ultimately support the implementation of targeted countermeasures to improve environmental quality, enhance community well-being, and help medical facilities prepare for patients during respiratory virus surges.

Similar forecasting projects have been conducted within the United States. CDC FluSight predicts flu-related hospitalizations each week based on past respiratory virus hospitalization data [9]. The Walgreens Respiratory Index ranks the top states for flu and COVID-19 activity as they occur by utilizing diagnostic testing data, respiratory illness treatments and over-the-counter purchases of flu or cold related products [10]. Kinsa Health detects the initial increase of infectious illness spread through their measured customer health metrics [11]. Outside the United States, projects have applied similar methodologies to comparable datasets. One such project uses clinical, demographic, air pollution, and meteorological factors to predict respiratory virus risks [12]. Another predicts daily respiratory disease counts using air pollution, climatic, and temporal factors [13].

Exploratory Data Analysis (EDA)

Data Collection & Understanding

The project uses the air quality dataset from the Open-Meteo API [2]. The spatial range covers a majority of Santa Clara County, centered around San Jose State University, with a temporal range from July 2023 to February 2026. To meet our goal of analyzing air quality in relation to health impacts, we supplemented this with the respiratory virus dataset from the California Department of Public Health database [3] [4] [5]. This secondary source provides an overview of viral activity, including positivity and severity rates for COVID-19, Influenza, and RSV. The spatial coverage includes the Bay Area region, while the temporal range matches the air quality dataset exactly.

Before selecting the respiratory virus dataset, we explored several options for supplemental health data to merge with our air quality dataset. The most ideal candidate was the Electronic Surveillance System for the Early Notification of Community-Based Epidemics (ESSENCE) [14]. This is a real-time epidemiological monitoring tool in Santa Clara County that tracks total Emergency Department visits for specific chief complaints, such as asthma-like symptoms. While its temporal buckets align perfectly with our air quality dataset, this information is unfortunately restricted to affiliated healthcare facilities and is not available to the public.

We also evaluated the Emergency Department Volume and Capacity dataset [15], but found it unusable. The data is too coarse, as it is grouped by cities and years, and it lacks coverage for our required 2023 to 2026 window.

Finally, we considered using asthma-related signals from Google Trends [16] due to its accessibility, and configurable spatial and temporal filters. We ultimately decided against this source because search volume does not confirm clinical admissions or actual diagnoses, making it a poor representation of our research goals.

The datasets underwent multiple phases of Exploratory Data Analysis (EDA). Initially, air quality and respiratory virus features were analyzed individually. In the latter phases, they were aggregated, combined, and analyzed together.

Data Quality Assessment

Individual analysis

The air quality dataset showed that 49.91% of the data were missing methane (µg/m³) and carbon_dioxide (ppm). Central tendency analysis reveals that most of the primary metrics are highly right-skewed. This implies that pollutant levels are low most of the time but events like fires and traffic spikes or industrial activity occasionally result in high readings. Some gases like carbon_monoxide (µg/m³) and carbon_dioxide (ppm) show a high Interquartile Range (IQR). There is also high variability for uv_index (), dust (µg/m³), nitrogen_dioxide (µg/m³), and sulphur_dioxide (µg/m³). These values vary strongly with the weather, time of day, and human activities.

In the respiratory virus dataset, we found that features such as tests done, admissions, admissions rates, and admission levels across all virus classes were missing 50 to 75% of data. We identified COV_POSITIVES, FLU_POSITIVES, and RSV_POSITIVES as the most reliable features due to their complete lack of null values and high variance. These features consistently show the strongest correlations with other variables and so we have initially designated them as our primary virus metrics. Our histogram analysis confirms there is sufficient distribution among the values to support meaningful data mining. Central tendency analysis reveals that most of these primary metrics are highly right-skewed. This confirms that virus activity remains low for most of the year and is interrupted only by spikes during peak seasons. From a data mining perspective this extreme skewness makes the mean an unreliable measure

of a typical week. Similar conclusions can be drawn from the variability of these features. The high coefficient of variation and elevated IQR values relative to Q1 and Q3 indicate that data points are spread across a vast range. This high volatility reflects a dataset containing distinct cycles of extreme highs and lows.
To ensure the respiratory virus dataset is symmetrical and comparable when merged with air quality dataset, transformations will be applied to stabilize feature variance and normalize the distribution.

Transform-then-aggregate analysis

Before the air quality and respiratory virus datasets are combined, transformations are applied to high-variance features to stabilize them. The transformations proved effective in reducing variance for most features. However, some features like the uv_index and dust (µg/m³) retained high variability despite the use of a complex transformation algorithm. We decided not to transform these features and instead wanted to see if aggregation before stitching the datasets together would stabilize them.

The air quality dataset must be aggregated into weekly buckets from the original hourly rows to match the respiratory virus dataset. After this process, we found that aggregation caused previously extremely skewed data like the uv_index and dust (µg/m³) to normalize but it also caused some stable features to gain skew. Moreover, certain transformations like log transformations limit the allowed transformations that can be performed later because they result in negative values that cannot be square-root transformed. Because of these findings, the team decided to revert the transformations and defer them until after the air quality dataset is aggregated.

Aggregate-then-transform analysis

Aggregating the air quality dataset features from hourly to weekly buckets is straightforward. Raw pollutants such as nitrogen_dioxide (µg/m³), carbon_monoxide (µg/m³) and ozone (µg/m³) are aggregated by their mean to indicate the typical exposure for the week. Indexes such as the uv_index () are aggregated by their maximum value to indicate the most significant hazard level for the week.

To identify the most effective way to normalize our data, we experimented with a set of reliable transformation methods including the square root, log, cube, and Yeo-Johnson transformations. Our findings simplified the selection process based on the severity of the skewness. Specifically, we found that the square root works best for moderately skewed data, log transformation is ideal for highly skewed data, and Yeo-Johnson transformation is most effective for extremely skewed data. The Yeo-Johnson method works by fitting the entire feature set and selecting an optimal exponent to smooth out the distribution [17]. These selections align with standard professional approaches for choosing data transformations [18].

Beyond these established standards, we evaluated whether each transformation successfully shifted the skewness into a specific target range. We aimed for a skewness value between -0.5

and below 0, or above 0 and 0.5, because we wanted the data to be approximately symmetric without achieving a perfect zero skew. This decision was intentional, as reaching exactly zero can introduce bias by over-smoothing. By maintaining this slight skew, we ensure the model remains sensitive to real-world fluctuations while mitigating the risks of underfitting or overfitting. We also addressed discrete text values, like COV_TP_LEVEL, FLU_TP_LEVEL, and RSV_TP_LEVEL, by converting them into numeric values. This discretization allows these features to be correlated with other variables, ensuring they can be used effectively in our final research model.

Further analysis of the aggregated, transformed, and stitched dataset confirmed that all features are now stabilized. Their box plots are centered with very minimal outliers, and their histograms more closely resemble a bell curve.

Pattern Discovery

To evaluate the internal relationships within the dataset, a correlation analysis was conducted. This process involved identifying highly correlated feature pairs and subsequently examining their scatter plots to determine the specific nature, shape, and distribution of their relationship.

During our individual analysis, we observed that the air quality features us_aqi (USAQI), us_aqi_pm2_5 (USAQI), and us_aqi_pm10 (USAQI) exhibited strong linear relationships. Their scatterplots revealed data points following a remarkably narrow path with very little spread, which confirms that these environmental factors move in close synchronization with one another. Within the respiratory virus dataset, the primary metrics COV_POSITIVES, FLU_POSITIVES, and RSV_POSITIVES initially appeared to be highly reliable indicators of community health crisis severity. These features correlated heavily with class-specific admissions, admission rates, and deaths. However, despite these strong correlations, we eventually decided to drop them. This decision was driven by the quality assessment findings, which showed a large portion of the data was missing. More importantly, the results observed during individual analysis shifted significantly once the datasets were analyzed together, suggesting that these features did not maintain their predictive value in a combined context.

The combined analysis reveals a significantly different perspective. As shown in Figure 1, RSV_TP_LEVEL exhibited the highest correlation amount and count among all air quality and respiratory dataset features. It is positively correlated with carbon_monoxide (μg/m³) and nitrogen_dioxide (μg/m³), which are pollutants that typically peak during cold seasons due to increased fuel combustion for heating [19] [20]. Conversely, it is negatively correlated with ozone (μg/m³) and the uv_index (), both of which are indicators that remain high during hot seasons [21] [22].

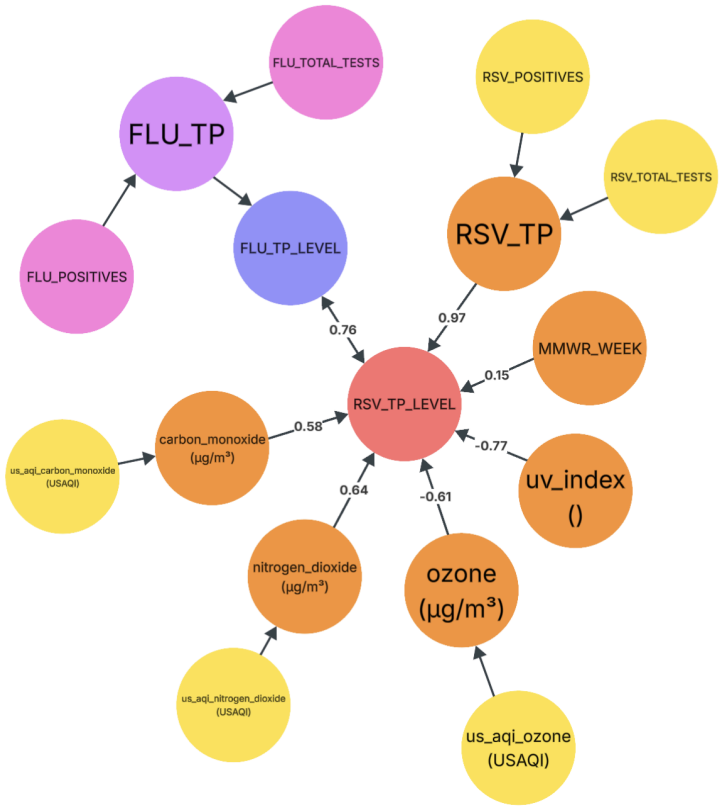

Figure 1 - Correlation diagram of RSV_TP_LEVEL and air quality features

While correlation and scatterplot analysis did not initially show a direct relationship between RSV_TP_LEVEL and MMWR_WEEK, a time series plot, as shown in Figure 2, illustrates that levels rise as winter arrives and fall as summer approaches.

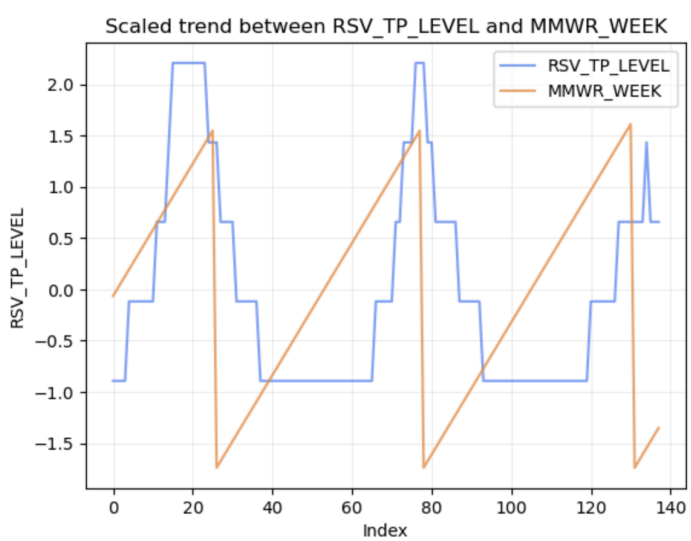

Figure 2 - Weekly distribution of RSV_TP_LEVEL

From these observations, we can deduce that RSV_TP_LEVEL is seasonal, which aligns with already established knowledge [23]. Consequently, it is correlated with seasonal air quality features, though this relationship is not necessarily causative. The seasonality of all these features is undeniably confirmed through hypothesis testing at a 99% confidence level, as detailed in Figure 3.

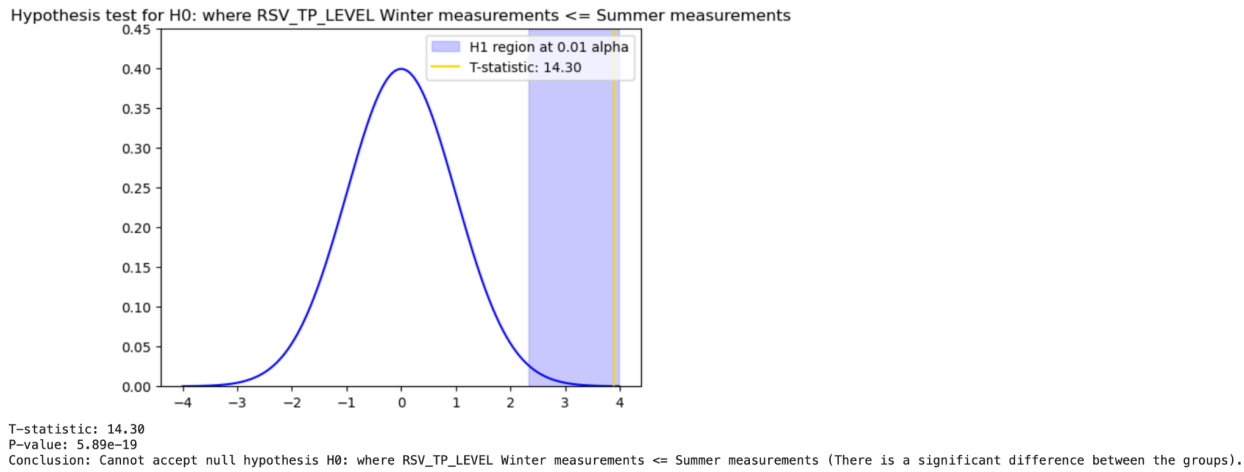

Figure 3 - Hypothesis Testing Confirming RSV_TP_LEVEL Seasonality at 99% Confidence

With this information at hand, we explored the possibility of air quality features being causative of RSV_TP_LEVEL fluctuations. Studies have indicated that exposure to nitrogen dioxide increases susceptibility to viruses [24] [25]. Additionally, research shows that exposure to sunlight or UV rays can deactivate viruses [22], and similar effects have been observed with ozone [26]. These findings require further research to be considered truly viable. For the purpose of this project, assuming seasonality is sufficient for our needs.

Data Preparation

As shown in Figure 1, RSV_TP_LEVEL is correlated directly or indirectly with positive cases, test positivity rates, test positivity levels, and total tests within the RSV and flu classes. This allows us to assume that its fluctuations correspond to trends in flu and other respiratory viruses in general. Additionally, this feature is more universal because it represents levels of positivity rather than a continuous rate, providing the same essential information as RSV test positivity rate in a simpler manner. Consequently, RSV_TP_LEVEL is the best candidate for our primary dependent feature, supported by MMWR_WEEK, carbon_monoxide (µg/m³), nitrogen_dioxide (µg/m³), ozone (µg/m³), and the uv_index () as independent features.

Data preparation is straightforward. As shown in Figure 4, the steps performed during the EDA are repeated. First, the respiratory virus dataset is filtered to include only data from the Bay Area region and the "all ages" age group. The air quality dataset is then aggregated before being stitched with the respiratory virus dataset. Finally, the combined dataset is sorted by

MMWR_YEAR and MMWR_WEEK to retain chronological order before proceeding to the next phase.

Dataset	Field	04-01-data-preparation Pick features	04-01-data-preparation Pick features	04-01-data-preparation Pick features	04-01-data-preparation Pick features	04-03-data-mining Stabilize data
Respiratory Virus	RSV_TP_LEVEL	None	None	None	None	Discretized
Air Quality	nitrogen_dioxide (µg/m³)	None	Aggregated (Mean)	None	None	Sqrt transformed
Air Quality	carbon_monoxide (µg/m³)	None	Aggregated (Mean)	None	None	Log transformed
Air Quality	uv_index ()	None	Aggregated (Max)	None	None	None
Air Quality	ozone (µg/m³)	None	Aggregated (Mean)	None	None	None
Respiratory Virus	MMWR_WEEK	None	None	Sorted by	None	None
Respiratory Virus	MMWR_YEAR	None	None	Sorted by	Dropped	
Respiratory Virus	SEASON	None	None	None	Dropped	
Respiratory Virus	WEEKENDING	None	None	None	Dropped	
Respiratory Virus	AGE_GRP	Filtered by	Dropped			
Respiratory Virus	RPHO_REGION	Filtered by	Dropped			

Figure 4: Transformation History of Features for Data Mining

Modeling

We used Random Forest Classifier as our main algorithms mainly because it handles small datasets well and significantly reduces overfitting. Random Forest has shown its ability to capture the non parametrics relationship between features and it is a reliable way to model this dataset. Feature Importance was derived from this algorithm. Also, Logistic Regression was used as a comparison tool to Random Forest; as well as, we want to see if there are some linear relationship between features or not. Furthermore, Stratified K-Fold cross validation with n = 4 was used in this project because we have only 138 samples. We used this Stratified K-Fold cross validation because it provided a more stable evaluation and worked well with a small and imbalanced dataset. We used Standard Scaler as the transformation tool to center the values between 0 and standard deviation of 1. The reason we use Standard Scaler is because some features have a very big difference in value. Thus, we transformed them to make sure the model converged and fair comparison.

Evaluation

The mean accuracy across the training and evaluation phases is 64.88% while the performance on the final test set reached a higher mark of 69.05%. While these numbers might initially appear modest, analyzing the confusion matrix in Figure 5 reveals that the model holds a deep understanding of the dataset. The predictions follow a clear diagonal trend which proves that when the model is wrong it usually only misses by a single category rather than making wild or illogical guesses. You can see that the accuracy is much higher for the low and very low levels which is a common result of having a small and imbalanced dataset. This performance is a sign that the model is struggling to generalize the rare peaks because it has not seen enough examples during training.

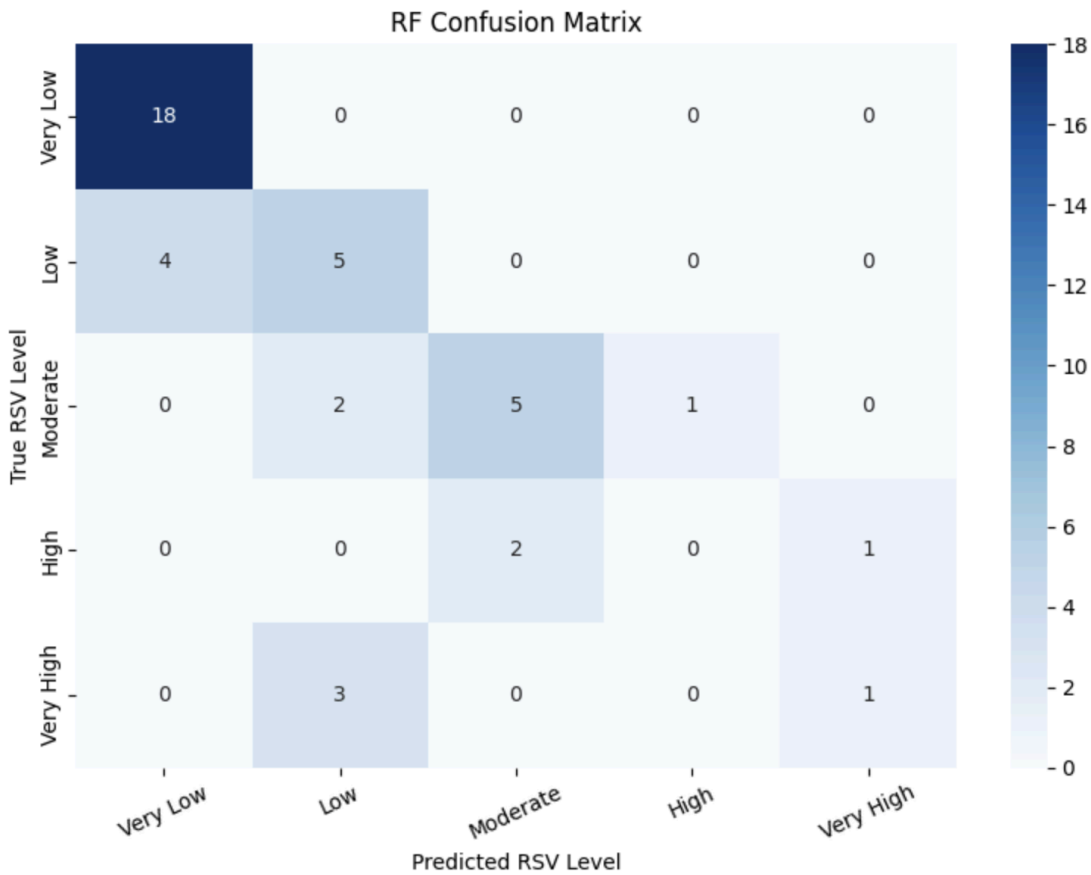

Figure 5: Random Forest confusion matrix

The solution to this limitation is to gather more training data to provide a wider variety of environmental scenarios. Despite the extreme noise in air quality sensors the model maintains a strong balance between identifying true cases and avoiding false alarms as shown by the F1 score of 0.66 in Figure 6. This balance is especially impressive given the class imbalance where low risk weeks far outnumber the high risk ones. The decision to use a Random Forest was validated because it outperformed the Logistic Regression in every metric which suggests that the links between air pollution and virus levels are complex and non linear.

Metrics	Random Forest	Logistic Regression	RF - LG
Accuracy	0.6905	0.6190	0.0715
F1 Score	0.6572	0.6007	0.0565
Precision	0.6415	0.6175	0.024
Recall	0.6904	0.6190	0.0714

Figure 6: Random Forest and Logistic Regression metric comparison

The Random Forest also provided vital insights into which environmental factors matter most for your predictions as shown in Figure 7. The UV index emerged as a primary driver with a significant importance score which confirms its role in influencing respiratory health trends. The epidemiological week closely followed in importance because it captures the essential seasonal patterns that a linear model might overlook. While nitrogen dioxide and ozone contributed less to the overall prediction they still provided a meaningful reduction in errors which proves that a multi sensor approach is necessary for a usable and robust model.

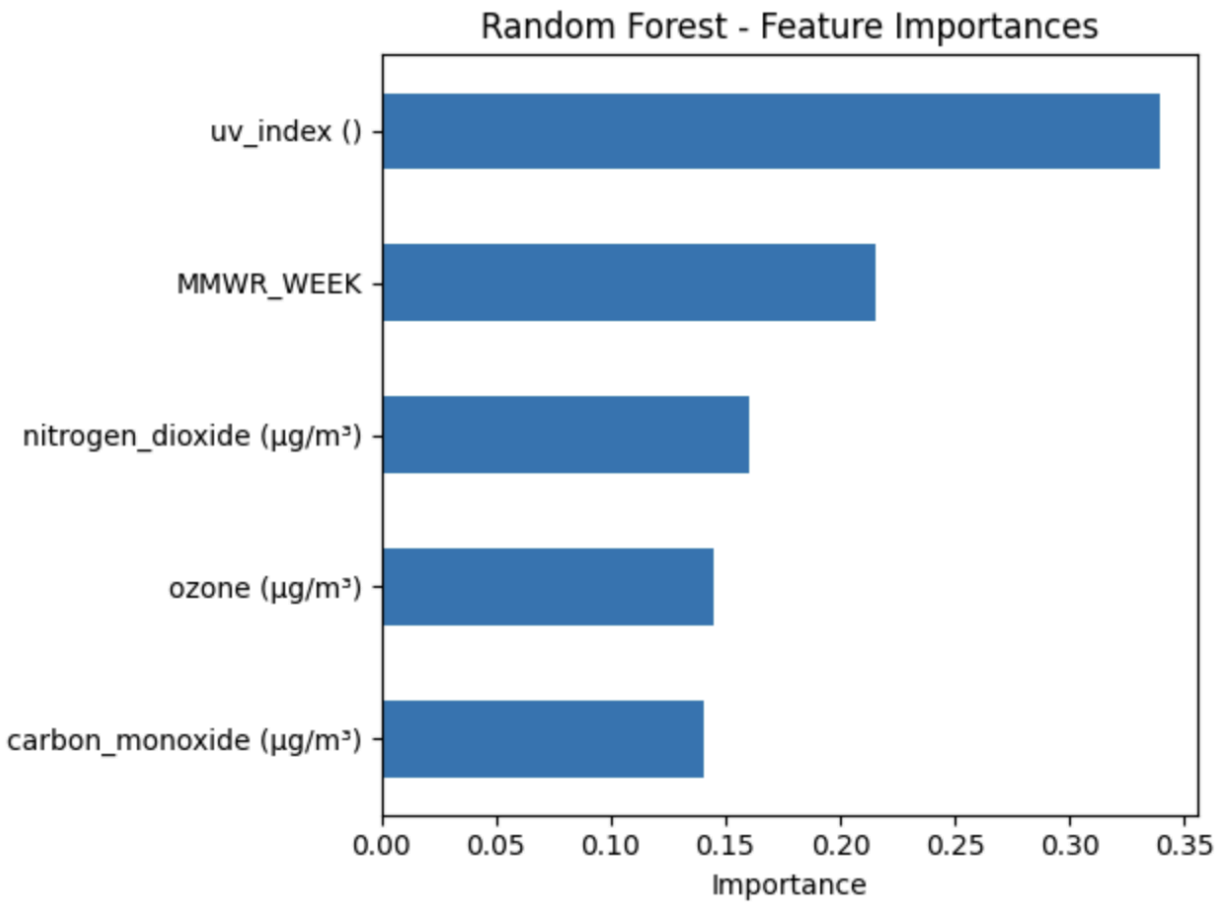

Figure 7: Random Forest Feature Importances

Deployment

The project's goal is to deploy a cloud based web dashboard that predicts respiratory virus test positivity level based on the epidemiological week and year. The application pulls data from the Open-Meteo API and feeds it into the model to generate predictions. These results are cached to ensure faster response times for repeat queries. To maintain high standards of reliability the system regularly compares its cached predictions with data from the California Department of Public Health. Additionally the model is designed to retrain and evaluate itself automatically whenever new data becomes available.

While other academic responsibilities limited our progress in the final stages we are proud of the foundational work we completed. This project establishes a framework for using air quality signals to predict respiratory health trends across the United States. It provides mathematical proof of the correlation between pollutants and viruses which serves as a resource for future clinical research into causation. Medical facilities can utilize these insights to reallocate resources ahead of expected surges to improve patient care. Ultimately if causation is proven these findings could justify public policy shifts toward renewable energy and electric appliances to help combat virus mortality.

Conclusion

We completed this project by following the CRISP-DM methodology for data mining which includes business understanding, exploratory data analysis, data preparation, modeling, evaluation, and deployment. While this framework ensured a smooth workflow we encountered several challenges that required adaptation.

The most significant hurdle was data availability. Although air quality datasets are abundant, finding a compatible respiratory virus dataset was difficult because strict patient privacy regulations prevent detailed health records from being made public. Consequently the available health data is very coarse grained to protect patient identities. This creates a mismatch with the fine granularity of the air quality data which makes precise temporal alignment a complex task.
Another challenge involved the varying results produced by different approaches to exploratory data analysis. During our initial pass correlation analysis suggested that particulate matter was the most significant feature due to its strong relationship with other environmental variables.

However a subsequent analysis revealed a different perspective. We found that while combustion gases do not correlate strongly with other environmental features they still maintain a notable correlation with the respiratory virus dataset itself. These conflicting signals highlight the complexity of selecting the right features for an accurate predictive model.

If administrative processing and time constraints are removed, the project would benefit significantly from utilizing the ESSENCE [14] dataset. Its daily temporal granularity aligns more closely with the air quality data, which would enable even more accurate predictions on a daily scale. Furthermore, integrating an air quality prediction model would allow for the forecasting of future respiratory virus test positivity levels, mirroring the methodologies of the related works cited in this study. Finally, incorporating location based features would facilitate specific predictions for individual cities, counties, or states. This level of detail would provide substantial value to the healthcare industry throughout the United States. Whether this requires a fundamental change to the architecture or simply training separate models for each geographic unit will be determined through future research.

One significant insight we gained through this project is the vital importance of domain knowledge. If we had possessed deep expertise from the start, we would have recognized that

pollutants resulting from combustion have a stronger correlation with respiratory virus test positivity levels than particulate matter does. This experience is comparable to developing financial transaction software where understanding the underlying business logic makes contributing to the application a seamless process. While domain knowledge is incredibly valuable, it is not strictly mandatory in the field of data mining. The very definition of the data mining process involves discovering hidden patterns within datasets, which is exactly what we achieved despite our initial lack of specialized domain expertise. If you are interested in the entire process we followed, please visit the project GitHub repository [1].

Definition of terms

methane (µg/m³) - A potent greenhouse gas that increases ozone levels.

carbon_dioxide (ppm) - Colorless gas produced by burning fossil fuels. Not harmful outdoors.

carbon_monoxide (µg/m³) - Odorless gas from combustion, reduces oxygen delivery to organs.

uv_index () - Measurement of the intensity of ultraviolet radiation from the sun, high levels can cause sunburn and skin cancer.z

dust (µg/m³) - Dust particles in the air that can cause coughing, wheezing, and asthma attacks.

nitrogen_dioxide (µg/m³) - Vehicle emissions can cause airway inflammation.

sulphur_dioxide (µg/m³) - Produced by burning fossil fuels, it constricts airways and is particularly dangerous for people with asthma.

COV_POSITIVES - Count of positive COVID-19 lab tests for the week.

FLU_POSITIVES - Count of positive Influenza lab tests for the week.

RSV_POSITIVES - Count of positive RSV lab tests for the week.

COV_DEATHS - Number of COVID-19 associated deaths each week.

FLU_DEATHS - Number of influenza-associated deaths each week.

RSV_DEATHS - Number of RSV-associated deaths each week.

ozone (µg/m³) - Created by chemical reactions between oxides of nitrogen and sunlight, it is a powerful lung irritant that can trigger asthma.

COV_TP_LEVEL - Positivity level based on 5 seasons of COVID-19 data.

FLU_TP_LEVEL - Positivity level based on 5 seasons of Influenza data.

RSV_TP_LEVEL - Positivity level based on 5 seasons of RSV data.

us_aqi (USAQI) - Air Quality Index is the overall standardized levels of health risk or air pollution.

us_aqi_pm2_5 (USAQI) - AQI for PM2.5 particles.

us_aqi_pm10 (USAQI) - AQI for PM10 particles.

MMWR_WEEK - Standardized CDC week number (1 to 52 or 53).

MMWR_YEAR - The calendar year associated with the reporting week.

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